Aerospace Structural Data Synthesis — Full Report

Confidential — Client Deliverable

Prepared from the recorded experiment artifacts on file.

Date: 2026-03-17

Champion: Empirical | Seed 11 | Size 5000 | Composite 0.9969

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Executive Summary

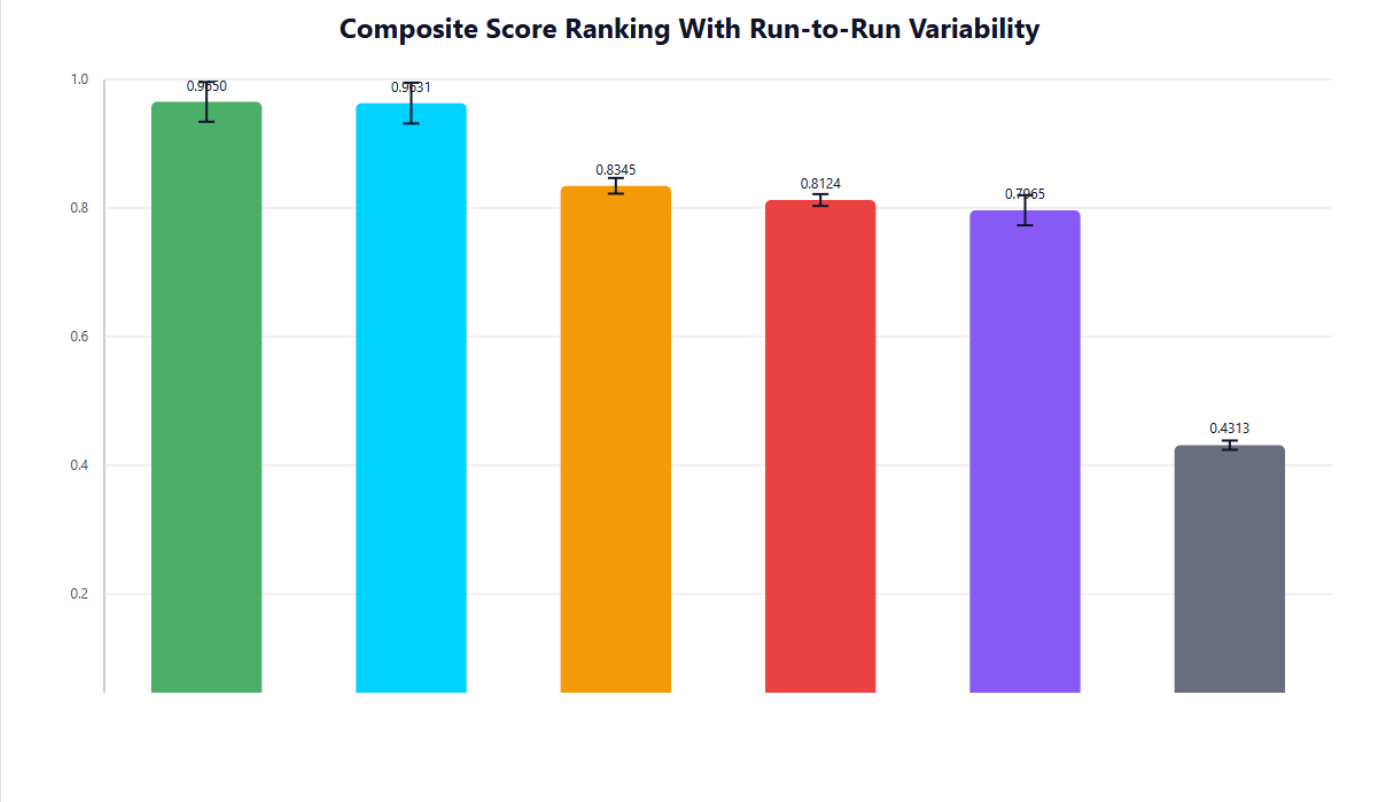
This report summarizes 72 recorded synthesis experiments across six model families for aerospace structural design data. The best observed run was Empirical with seed 11 and size 5000, reaching a composite score of 0.9969. KDE produced the strongest average score across all of its runs, but Empirical supplied the single best client-ready configuration. The recommended operational policy is to use Empirical or KDE at size 1000 or above.

Methodology

The pipeline profiled the original 300-row, 22-feature aerospace dataset, trained six model families over three seeds and four generation scales, evaluated four quality pillars, and then ranked every run using the composite score. This deliverable package was assembled strictly from those recorded artifacts without rerunning the experiment suite.

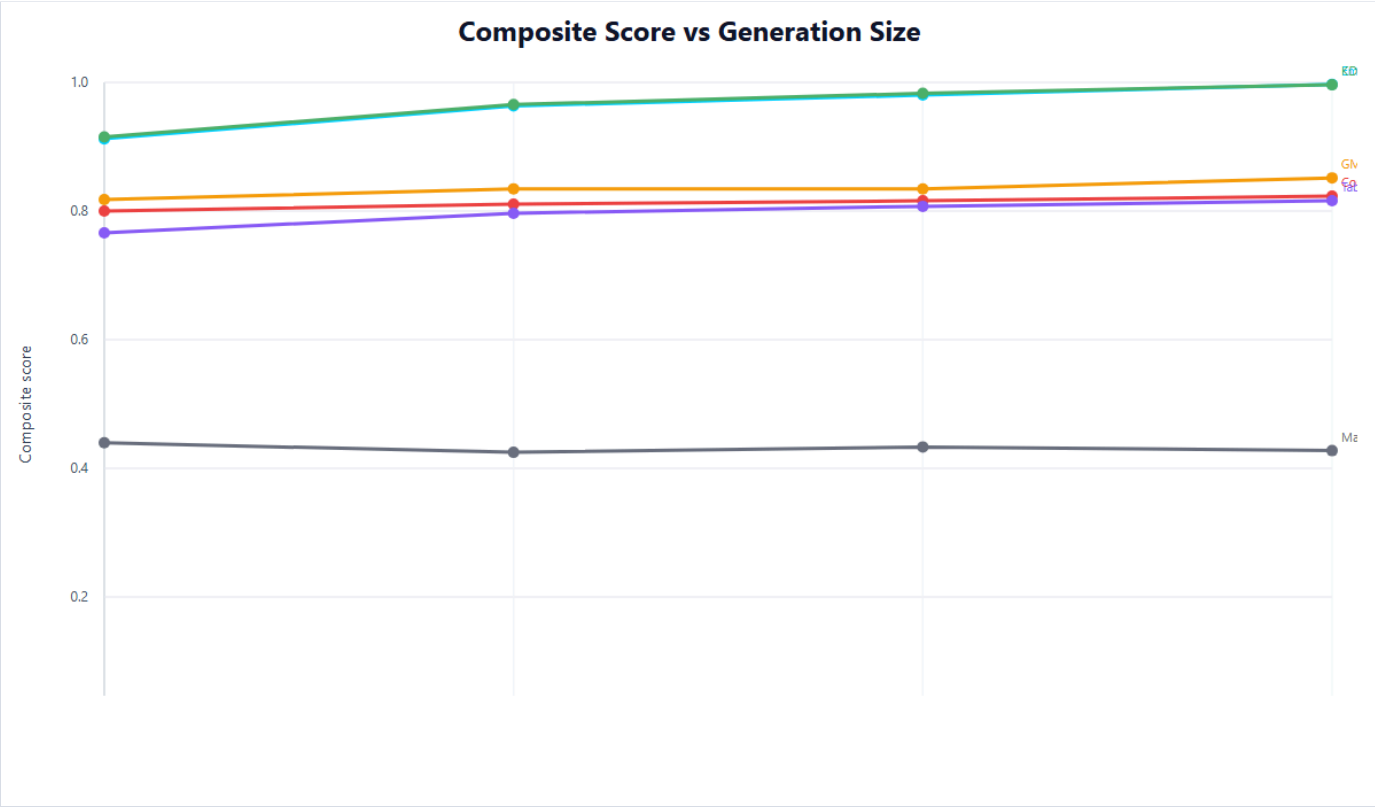
Model	Mean Composite	Mean Physics	Mean Stats	Mean TSTR	Mean Privacy
Empirical	0.9631	0.9911	0.9828	0.9050	0.9828
KDE	0.9650	0.9924	0.9828	0.9089	0.9828
GMM	0.8345	0.9897	0.9781	0.4019	0.9781
Copula	0.8124	0.9895	0.9669	0.3319	0.9669
TabuDiff	0.7965	0.9822	0.9107	0.3789	0.9107
MaskedPredictor	0.4313	0.9587	0.0000	0.4408	0.0000

Model Benchmarking Results

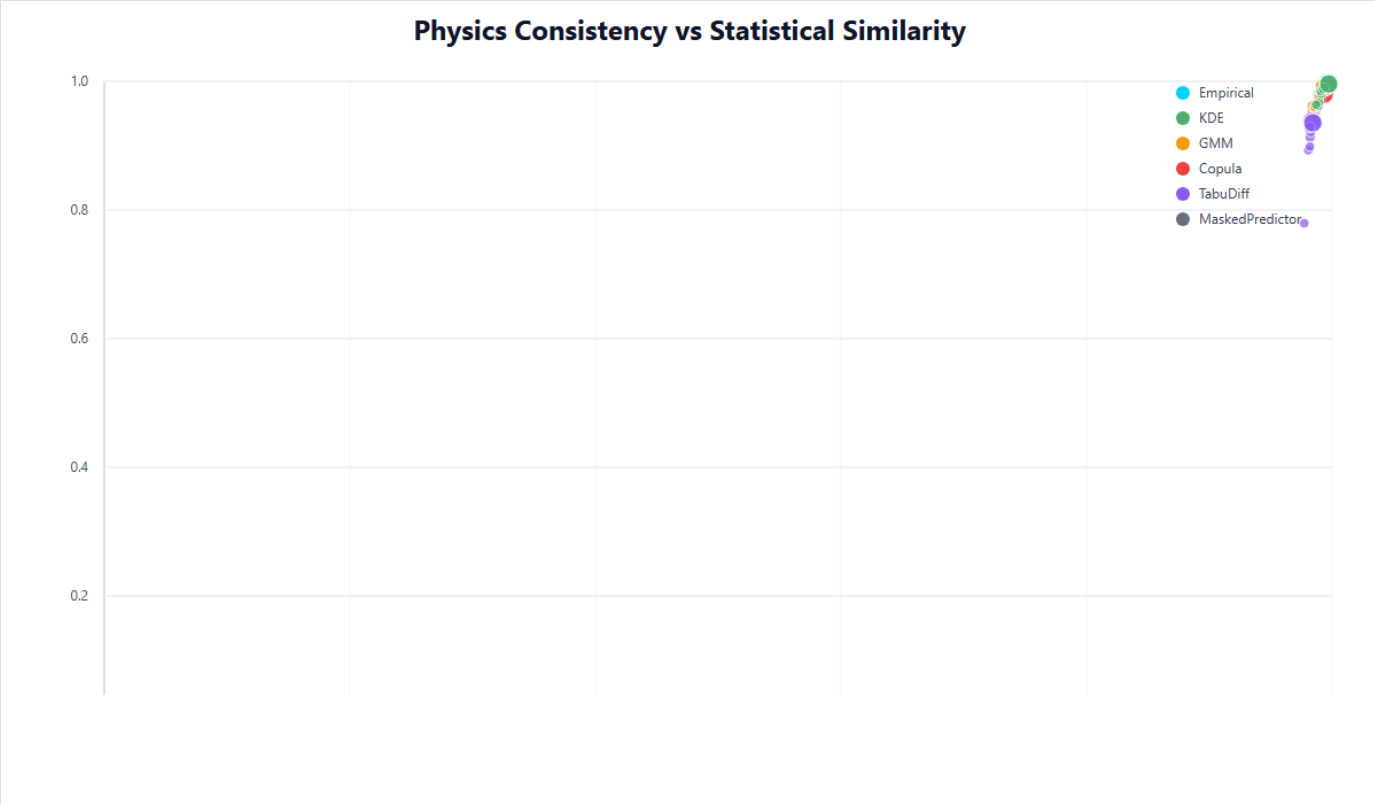


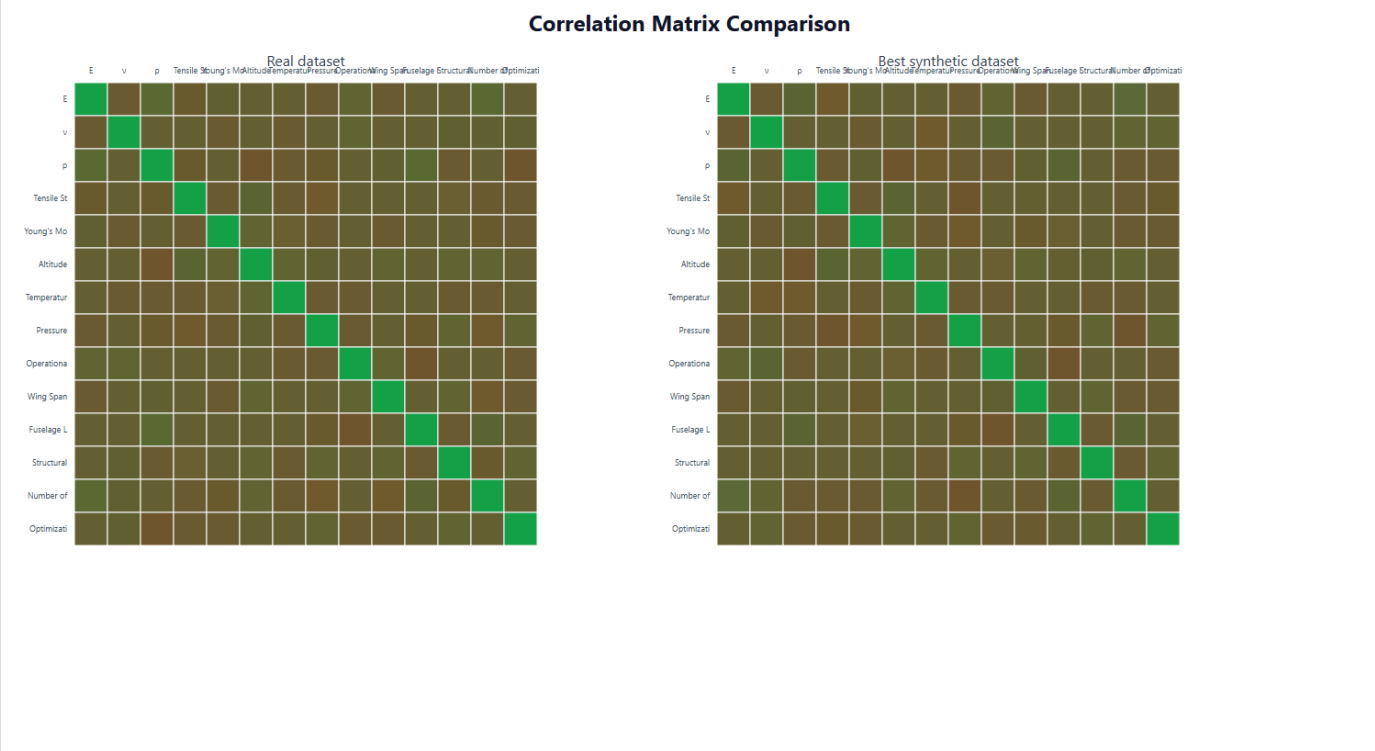
Mean Metrics Heatmap By Model

	Physics	Statistical	ML Utility	Privacy	Composite Score
Empirical	0.991	0.983	0.905	0.983	0.963
KDE	0.992	0.983	0.909	0.983	0.965
GMM	0.990	0.978	0.402	0.978	0.835
Copula	0.990	0.967	0.332	0.967	0.812
TabuDiff	0.982	0.911	0.379	0.911	0.797
skedPredictor	0.959	0.000	0.441	0.000	0.431

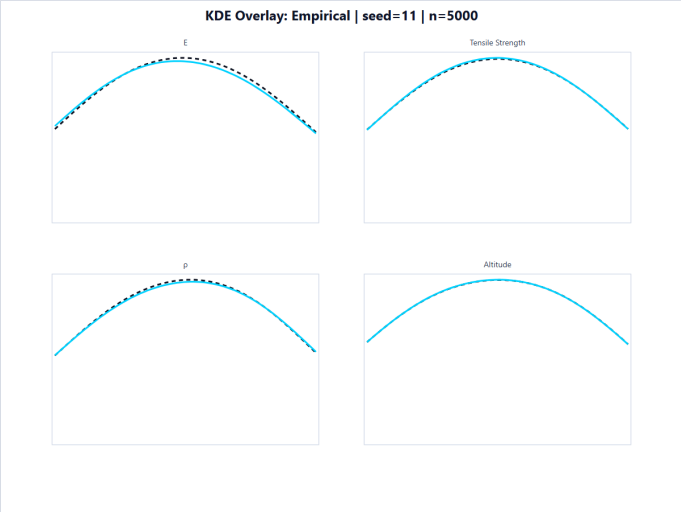

Composite Score vs Generation Size
Composite score
1.0
0.8
0.6
0.4
0.2

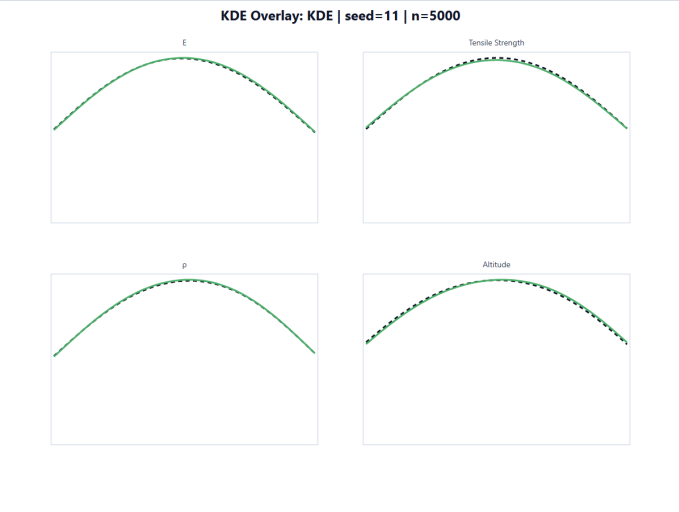
Statistical Analysis



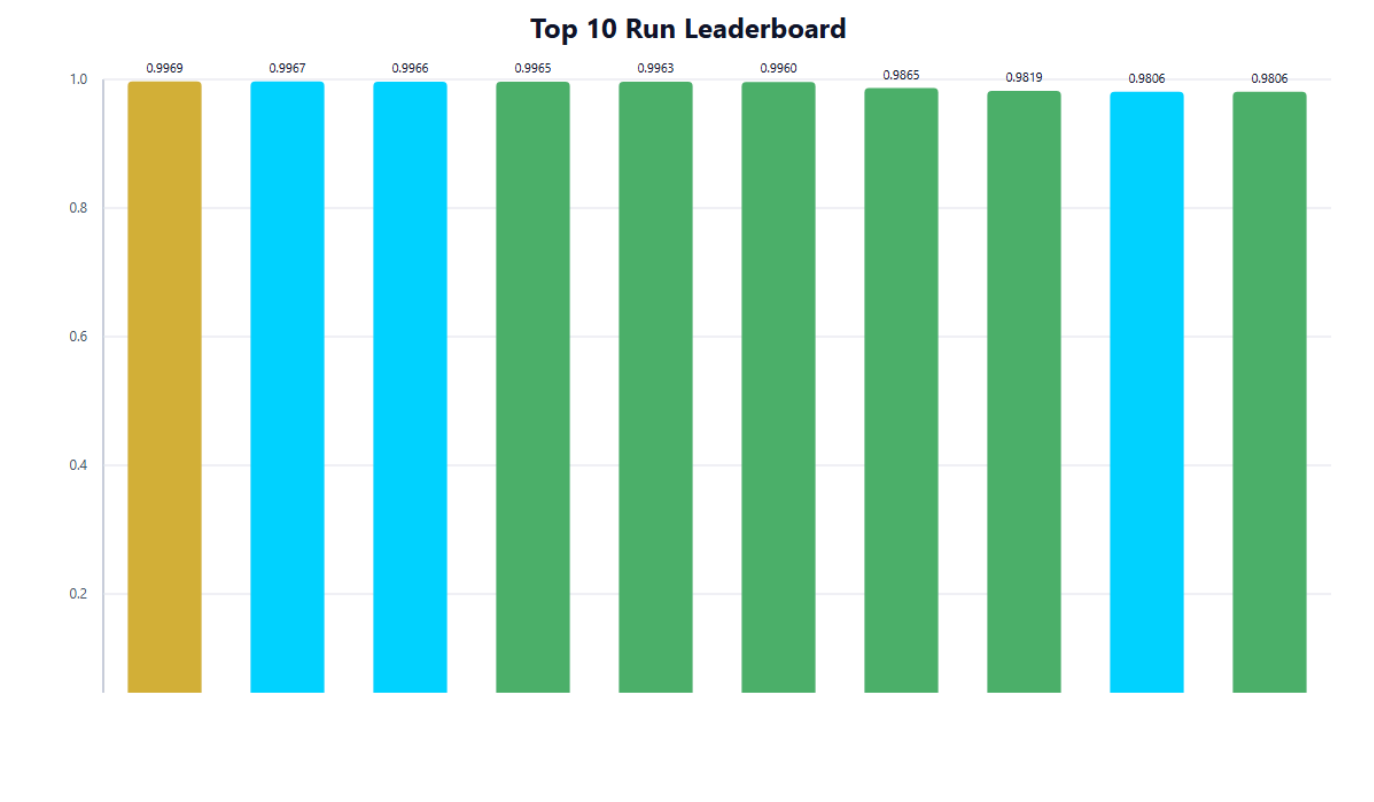


KDE Overlay Visuals





Top 10 Leaderboard



Rank	Model	Seed	Size	Composite
1	Empirical	11	5000	0.9969
2	Empirical	123	5000	0.9967
3	Empirical	42	5000	0.9966
4	KDE	123	5000	0.9965
5	KDE	11	5000	0.9963
6	KDE	42	5000	0.9960
7	KDE	123	1000	0.9865
8	KDE	42	1000	0.9819
9	Empirical	123	1000	0.9806
10	KDE	11	1000	0.9806

Best Synthetic Dataset Sample

Material Type	E (GPa)	ν (Poisson)	ρ (kg/m³)	Tensile Strength (MPa)	Young's Modulus	Structural Shape	Load Distribution	Altitude (m)	Temperatur (°C)
Titanium	126.50016901527832	0.3425330024107249	3674.758496339591	664.9957792545542	84.42230576682806	Rectangular	distributed	9402.14077546416	-53.646455
Aluminum	109.69766873153114	0.3180160347567902	2581.6527658101213	1121.0376109773708	53.73946761636116	Rectangular	uniform	16203.24088097327	-46.512067
Aluminum	162.6254959	0.3437990888237314	3551.899938834394	1017.1737632298413	80.00480642199898	Tapered	uniform	10712.328185207763	-48.087728
Carbon Fiber	186.7704624284862	0.3006367583799147	3965.998547	865.0213275297234	92.62167016322528	Rectangular	point load	13699.36069643242	-47.589738
Aluminum	126.24244126027872	0.3288802504631441	4171.673435526482	980.4782303	50.10844500858114	Rectangular	uniform	6793.918802491668	-40.409615
Carbon Fiber	199.32331858444863	0.3234672408279735	3641.359294	914.8748181950556	70.12820591	Rectangular	point load	9471.049509	-57.881655
Aluminum	115.29300684919107	0.3184380080271858	4125.936865372625	1214.1966605504322	91.94549697	Cylindrical	uniform	7274.916556316738	-57.921855
Titanium	114.58792068307096	0.3471798872992859	3548.066634951153	472.26818044919537	60.825413582155406	Cylindrical	uniform	12397.296734391202	-44.968096
Carbon Fiber	93.56137557852628	0.3163487031851667	4222.337339056994	514.4395642182785	54.43976358531143	Rectangular	point load	11032.83174127196	-41.884592
Titanium	73.32930465871681	0.3427274540188058	2534.934963035566	490.3231064566148	50.75393812627439	Tapered	distributed	5778.830183082483	-50.974139
Titanium	60.484640668248126	0.3257484657981025	1793.997006732732	335.2350290226964	60.156575490146906	Rectangular	point load	5086.0	-41.626859
Carbon Fiber	191.8542837494378	0.3037113248683276	1586.7272265085812	1313.9306437749235	78.21852152618891	Rectangular	point load	18824.23209280477	-58.520916
Aluminum	183.8906167180615	0.3076802525024759	2792.9898542986843	432.0151760907893	64.35188629950419	Cylindrical	point load	12449.436906372675	-50.047748
Titanium	83.94742701811707	0.3462976041384147	1620.5408343974364	1410.3053664746371	72.45802238788207	Rectangular	uniform	5974.708676638895	-40.830608
Aluminum	114.08455086521978	0.3285860677801487	2579.618700241688	1096.2366018135808	50.75744203920824	Cylindrical	point load	15770.722855352524	-46.777237

Conclusions & Recommendations

1. Use Empirical or KDE at size ≥ 1000 for production synthesis tasks.
2. Prefer Empirical as the default because it delivered the strongest single configuration with the lowest operational complexity.
3. Keep GMM and Copula as explanatory baselines, not primary delivery models.
4. Defer MaskedPredictor pending architecture overhaul.
5. Explore conditional and ensemble synthesis as the next phase of work.

Appendix A: Full 72-run Metrics

Model	Seed	Size	Physics	Stats	TSTR	Privacy	Composite
Copula	11	300	0.9835	0.9518	0.3067	0.9518	0.7959
Copula	11	600	0.9885	0.9644	0.3167	0.9644	0.8069
Copula	11	1000	0.9900	0.9729	0.3233	0.9729	0.8130
Copula	11	5000	0.9927	0.9818	0.3733	0.9818	0.8309
Copula	42	300	0.9865	0.9521	0.3267	0.9521	0.8030
Copula	42	600	0.9898	0.9665	0.3300	0.9665	0.8120
Copula	42	1000	0.9913	0.9711	0.3467	0.9711	0.8190
Copula	42	5000	0.9930	0.9806	0.3167	0.9806	0.8165
Copula	123	300	0.9860	0.9475	0.3267	0.9475	0.8009
Copula	123	600	0.9889	0.9628	0.3433	0.9628	0.8132
Copula	123	1000	0.9904	0.9710	0.3367	0.9710	0.8159
Copula	123	5000	0.9937	0.9798	0.3367	0.9798	0.8216
Empirical	11	300	0.9864	0.9662	0.7500	0.9662	0.9144
Empirical	11	600	0.9912	0.9835	0.9200	0.9835	0.9673
Empirical	11	1000	0.9920	0.9884	0.9633	0.9884	0.9806
Empirical	11	5000	0.9967	0.9979	1.0000	0.9979	0.9969
Empirical	42	300	0.9850	0.9591	0.7500	0.9591	0.9107
Empirical	42	600	0.9884	0.9810	0.9167	0.9810	0.9635
Empirical	42	1000	0.9914	0.9885	0.9600	0.9885	0.9794
Empirical	42	5000	0.9967	0.9973	1.0000	0.9973	0.9966
Empirical	123	300	0.9866	0.9632	0.7433	0.9632	0.9117
Empirical	123	600	0.9900	0.9802	0.8967	0.9802	0.9593
Empirical	123	1000	0.9923	0.9901	0.9600	0.9901	0.9806
Empirical	123	5000	0.9965	0.9979	1.0000	0.9979	0.9967
GMM	11	300	0.9835	0.9620	0.3700	0.9620	0.8157
GMM	11	600	0.9887	0.9742	0.4167	0.9742	0.8359
GMM	11	1000	0.9911	0.9827	0.3767	0.9827	0.8310
GMM	11	5000	0.9936	0.9917	0.4233	0.9917	0.8480
GMM	42	300	0.9863	0.9595	0.3833	0.9595	0.8200
GMM	42	600	0.9884	0.9784	0.4067	0.9784	0.8349
GMM	42	1000	0.9910	0.9832	0.3867	0.9832	0.8337
GMM	42	5000	0.9940	0.9921	0.4433	0.9921	0.8535
GMM	123	300	0.9857	0.9601	0.3733	0.9601	0.8173
GMM	123	600	0.9897	0.9774	0.3967	0.9774	0.8329
GMM	123	1000	0.9910	0.9845	0.4033	0.9845	0.8383
GMM	123	5000	0.9939	0.9913	0.4433	0.9913	0.8531
KDE	11	300	0.9886	0.9616	0.7500	0.9616	0.9142
KDE	11	600	0.9910	0.9822	0.9033	0.9822	0.9624
KDE	11	1000	0.9926	0.9875	0.9633	0.9875	0.9806
KDE	11	5000	0.9971	0.9957	1.0000	0.9957	0.9963
KDE	42	300	0.9894	0.9672	0.7600	0.9672	0.9195
KDE	42	600	0.9913	0.9809	0.9067	0.9809	0.9629
KDE	42	1000	0.9935	0.9891	0.9633	0.9891	0.9819
KDE	42	5000	0.9969	0.9955	1.0000	0.9955	0.9960
KDE	123	300	0.9870	0.9638	0.7433	0.9638	0.9122
KDE	123	600	0.9909	0.9854	0.9333	0.9854	0.9712
KDE	123	1000	0.9930	0.9889	0.9833	0.9889	0.9865
KDE	123	5000	0.9973	0.9960	1.0000	0.9960	0.9965

Model	Seed	Size	Physics	Stats	TSTR	Privacy	Composite
MaskedPredictor	11	300	0.9592	0.0000	0.4733	0.0000	0.4398
MaskedPredictor	11	600	0.9571	0.0000	0.3867	0.0000	0.4167
MaskedPredictor	11	1000	0.9577	0.0000	0.4333	0.0000	0.4287
MaskedPredictor	11	5000	0.9582	0.0000	0.4500	0.0000	0.4332
MaskedPredictor	42	300	0.9592	0.0000	0.4733	0.0000	0.4398
MaskedPredictor	42	600	0.9599	0.0000	0.4200	0.0000	0.4270
MaskedPredictor	42	1000	0.9606	0.0000	0.4667	0.0000	0.4391
MaskedPredictor	42	5000	0.9590	0.0000	0.4100	0.0000	0.4238
MaskedPredictor	123	300	0.9592	0.0000	0.4733	0.0000	0.4398
MaskedPredictor	123	600	0.9571	0.0000	0.4433	0.0000	0.4308
MaskedPredictor	123	1000	0.9587	0.0000	0.4433	0.0000	0.4319
MaskedPredictor	123	5000	0.9590	0.0000	0.4167	0.0000	0.4254
TabuDiff	11	300	0.9806	0.8931	0.3633	0.8931	0.7845
TabuDiff	11	600	0.9821	0.9132	0.3700	0.9132	0.7953
TabuDiff	11	1000	0.9830	0.9315	0.3800	0.9315	0.8057
TabuDiff	11	5000	0.9836	0.9385	0.4367	0.9385	0.8231
TabuDiff	42	300	0.9773	0.7793	0.3300	0.7793	0.7284
TabuDiff	42	600	0.9817	0.9251	0.3733	0.9251	0.8006
TabuDiff	42	1000	0.9828	0.9285	0.3767	0.9285	0.8035
TabuDiff	42	5000	0.9842	0.9356	0.3967	0.9356	0.8123
TabuDiff	123	300	0.9819	0.8987	0.3533	0.8987	0.7851
TabuDiff	123	600	0.9826	0.9215	0.3500	0.9215	0.7939
TabuDiff	123	1000	0.9817	0.9286	0.4167	0.9286	0.8128
TabuDiff	123	5000	0.9844	0.9355	0.4000	0.9355	0.8133